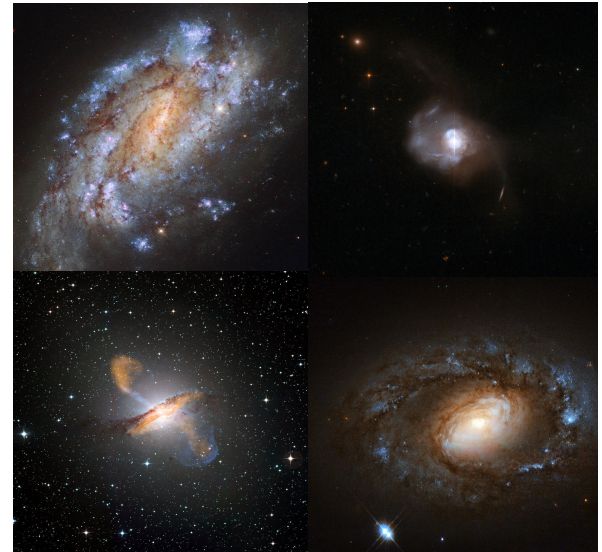


Improving Machine-learning Classifiers for Low Redshift Emission-line Galaxies

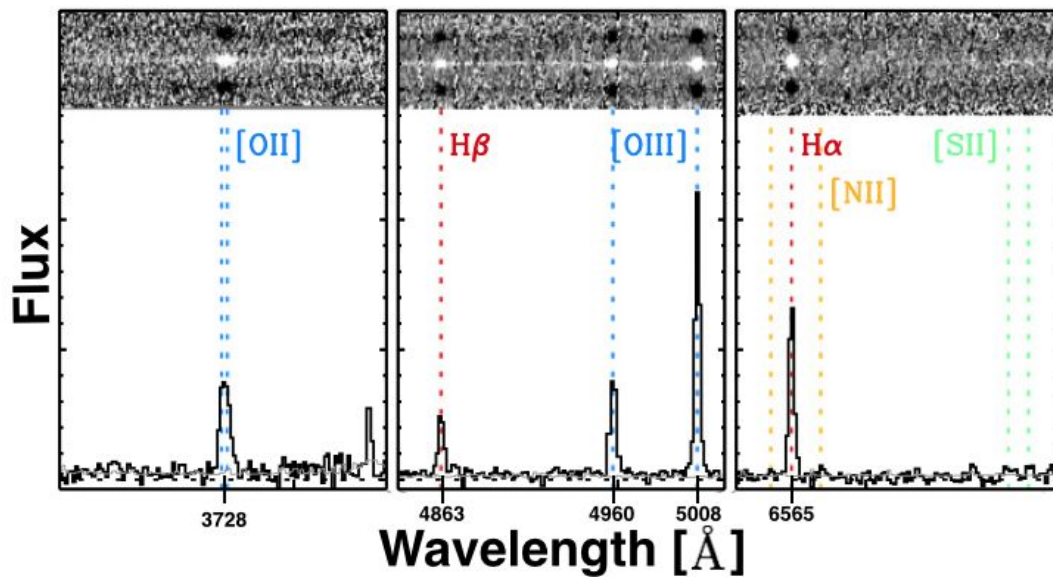
Final Project - Deep Learning in Physics
Lorenzo Cesario, Felix Schulz, Csilla Tijssen



The Paper

Zhang et al. (2019): Machine-learning Classifiers for Intermediate Redshift Emission-line Galaxies

- Star-forming (SF), Active Galactic Nuclei (AGN), Composites, Low-ionization narrow emission-line regions (LINERs)
- Optical diagnostic diagrams



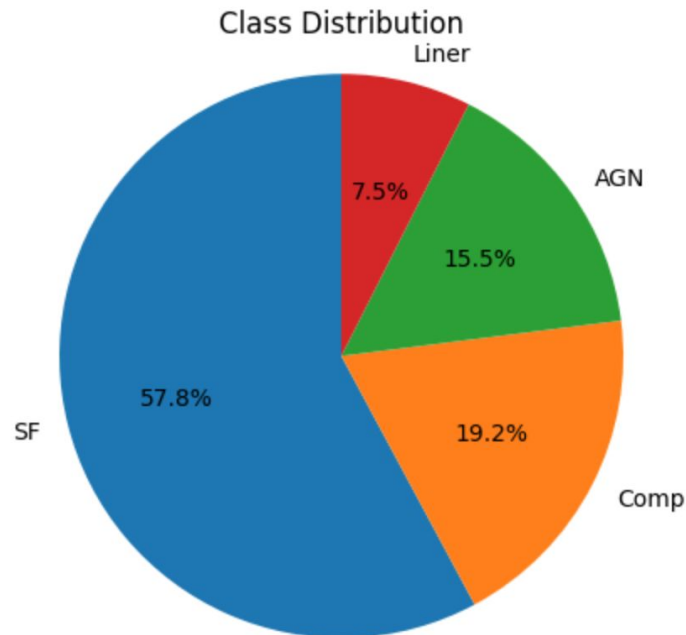
Kriek et al. (2015)

The Models

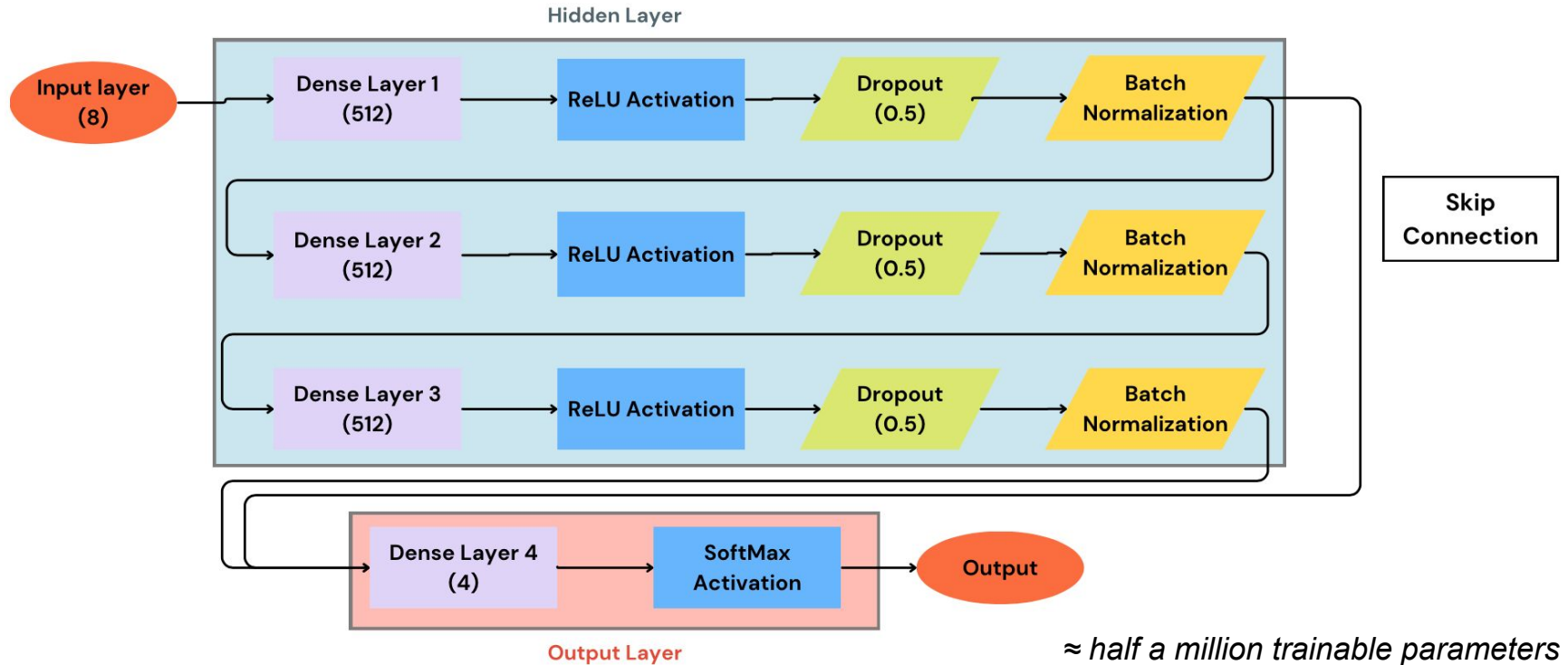
- Input features:
 - $[\text{O II}]/\text{H}\beta$, $[\text{O III}]/\text{H}\beta$, $[\text{O III}]$ line width
 - Stellar velocity dispersion
 - Colors: $u - g$, $g - r$, $r - i$, $i - z$
- k -nearest neighbor (kNN), support vector classifier (SVC), random forest (RF), multilayer perceptron (MLP) neural network
- ROC curves $\rightarrow$ AUC; confusion matrix and balanced accuracy

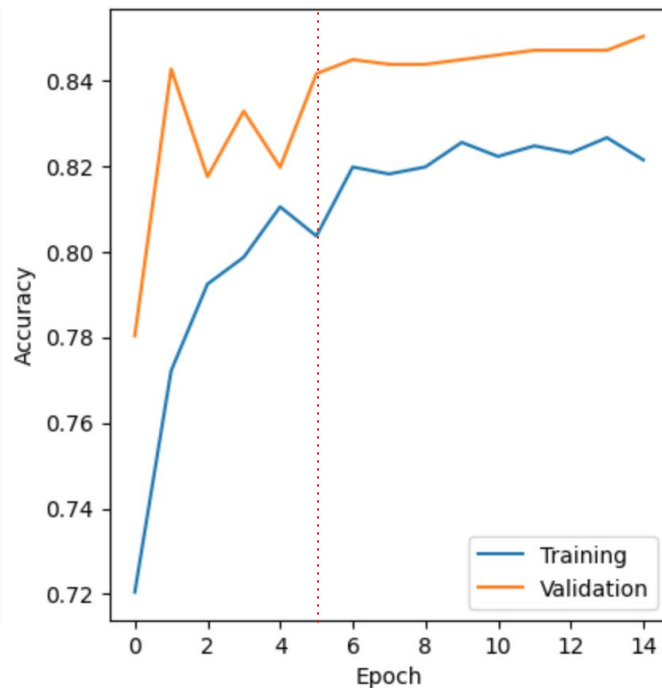
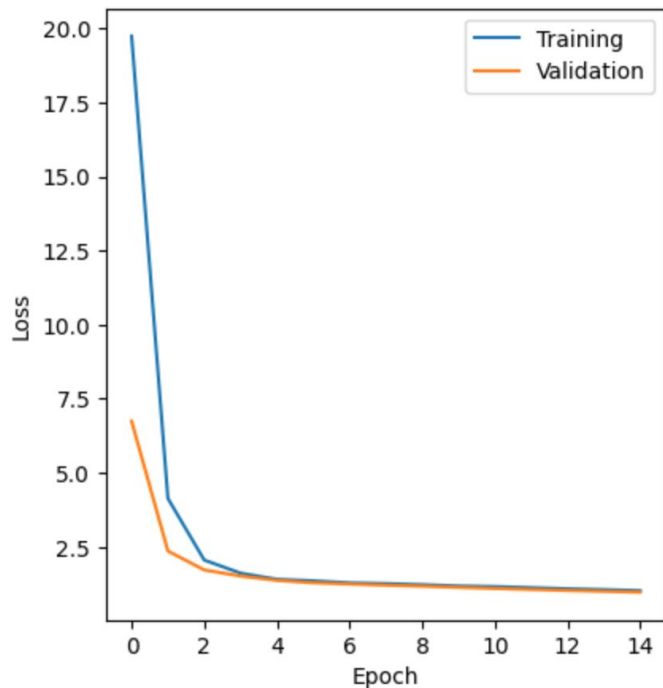
The Data

- Imbalanced class problem
- From SDSS eBOSS (spectral data)
- Filtering → 28 869 galaxies
- 70/30 + 6-fold splitting



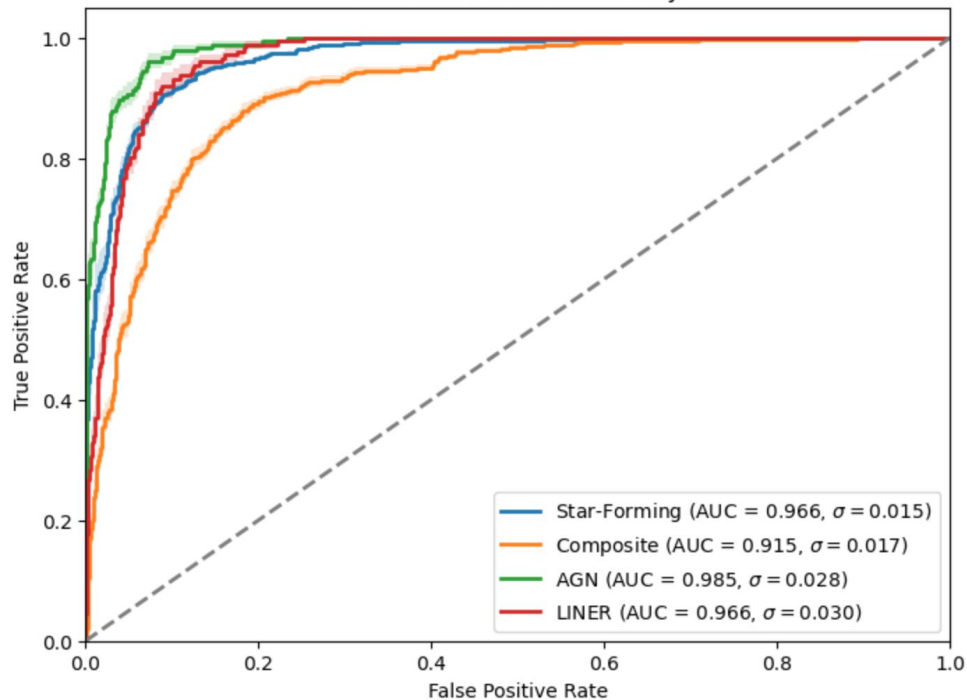
Our Neural Network





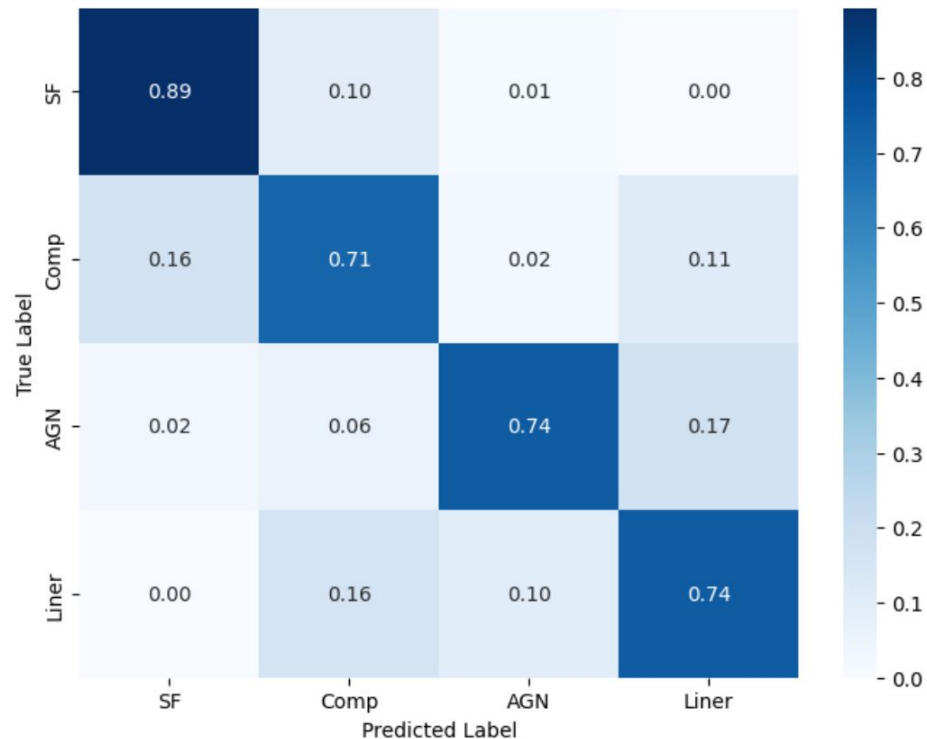
- Standard Scaler on data
- L2 regularizer at every layer
 - **generalizable** patterns
 - discourage **large weights**
- Learning rate switch at 5th epoch
 - $10^{-3} \rightarrow 10^{-5}$

ROC Curve with Uncertainty



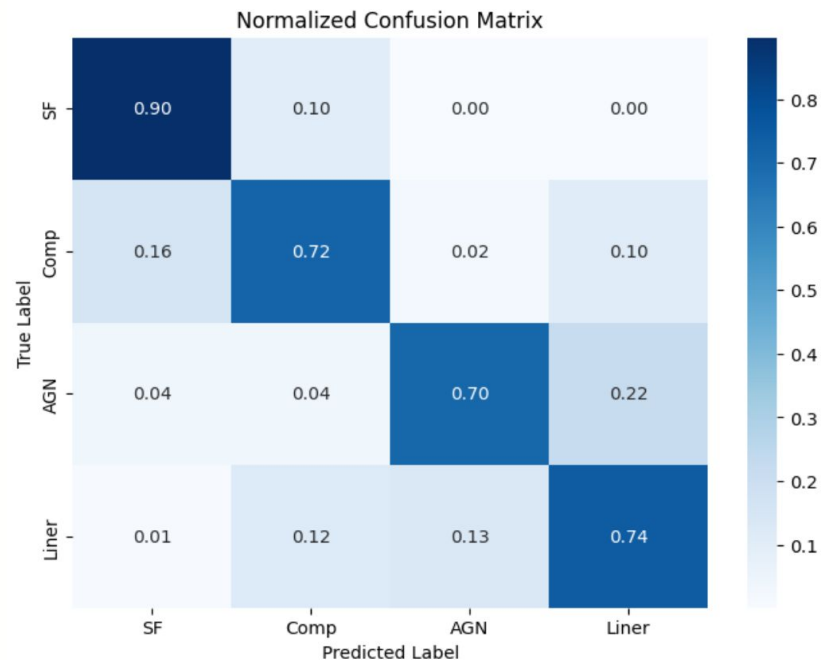
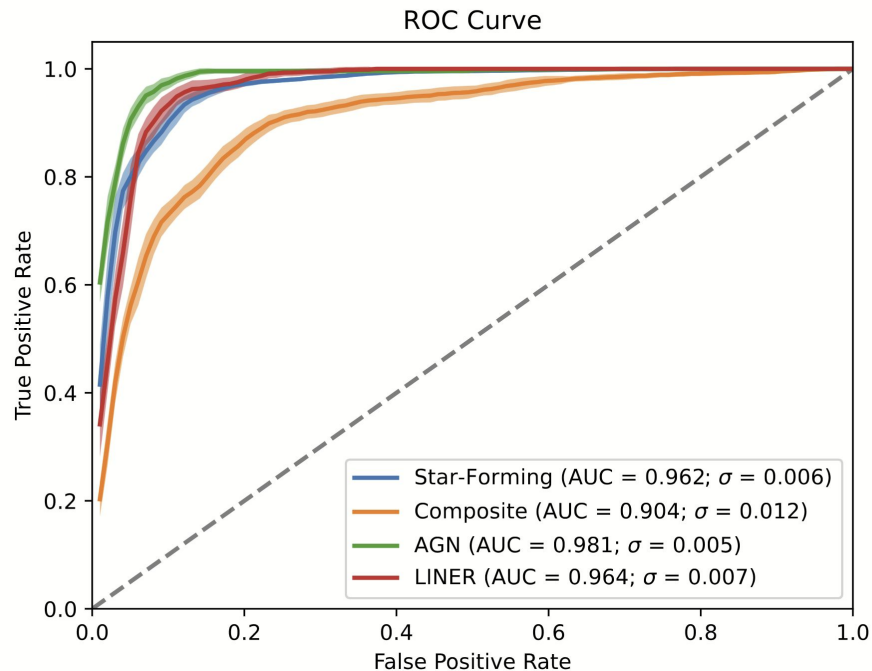
- Similar AUC for **Star-Forming** class
- Higher AUC for all other galaxies
- Uncertainty calculated with **Monte Carlo Dropout** method
 - 50 samples

Normalized Confusion Matrix



Balanced accuracy $\approx 77\%$

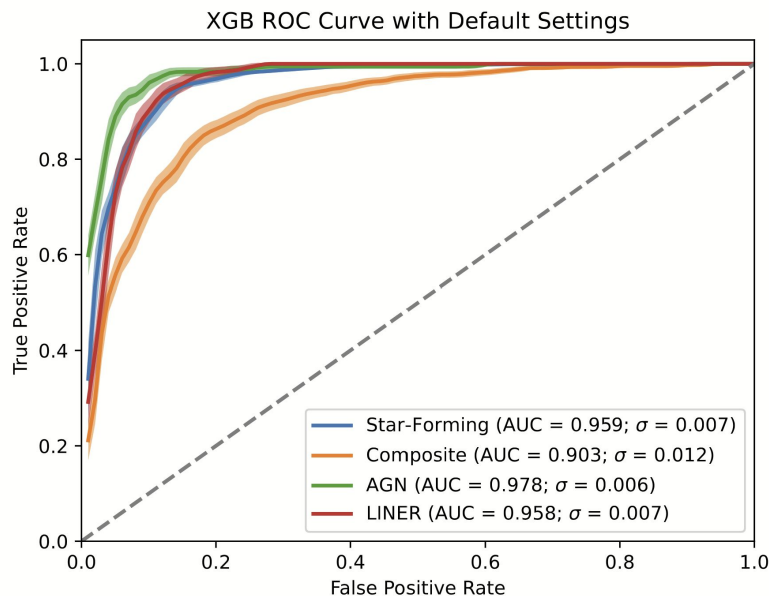
Random Forest

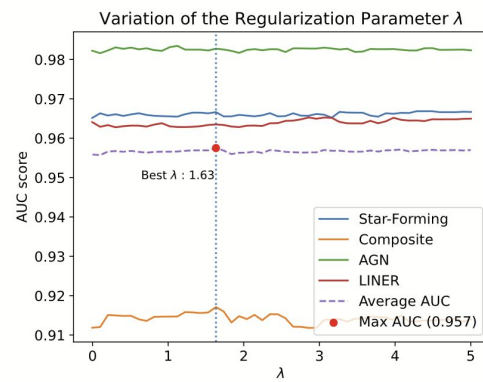
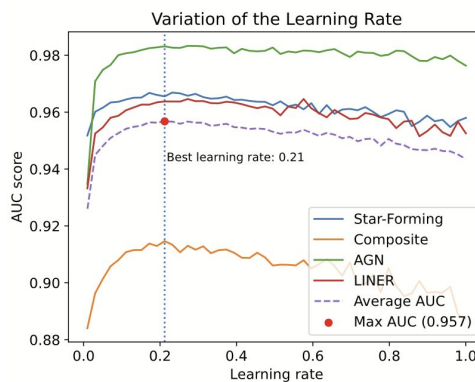
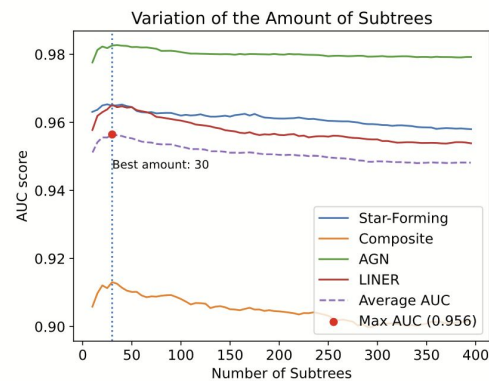
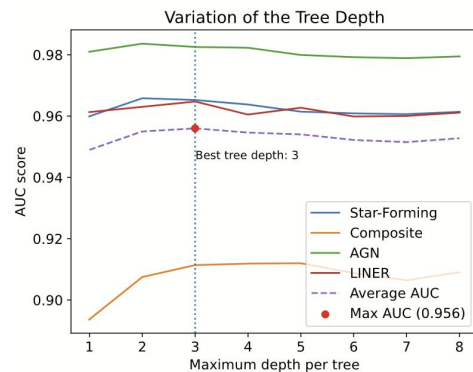


- Similar AUC for SF
- Lower AUC for Composites
- Higher AUC for AGN and LINERs
- Uncertainty calculated by **bootstrap**
 - Visualize model's **sensitivity** to "new data"
 - Do predictions vary?

Gradient-Boosted Model

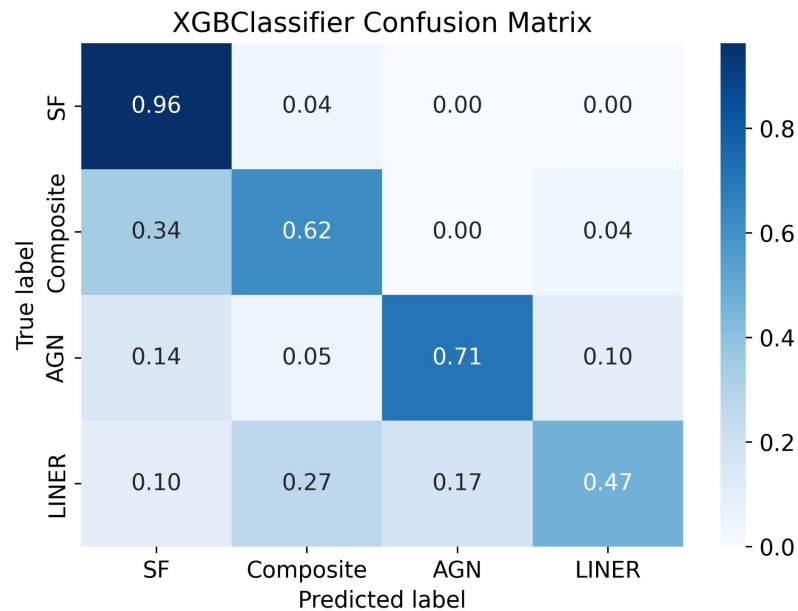
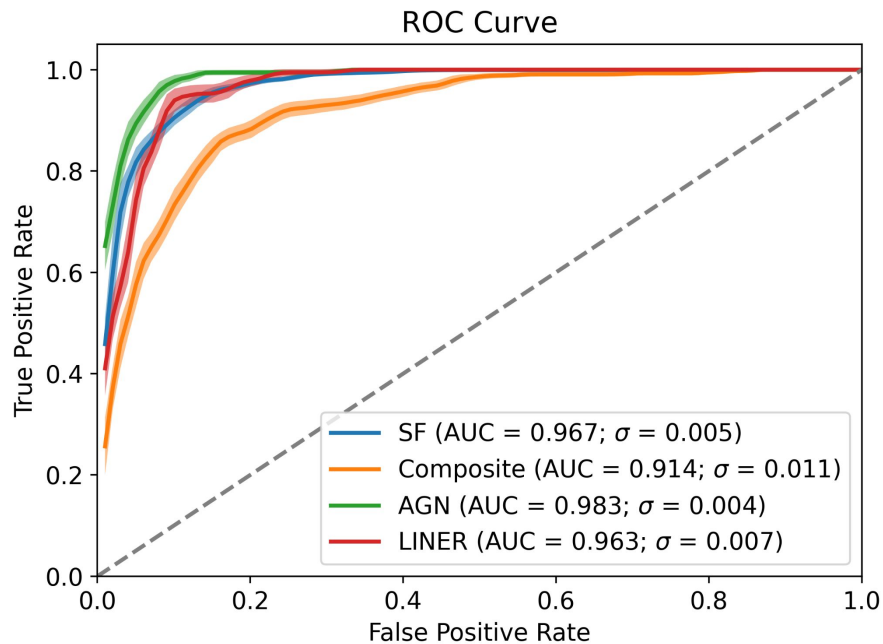
- Using XGBClassifier from the XGBoost library
- Parameters we will vary:
 - max_depth (default 3)
 - n_estimators (default 100)
 - learning_rate (default 0.3)
 - reg_lambda (default 1)



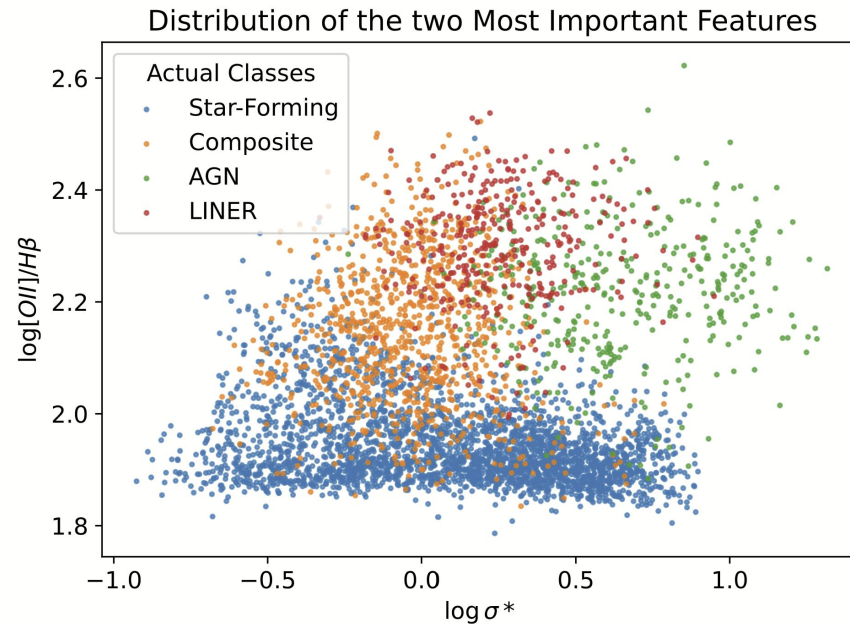
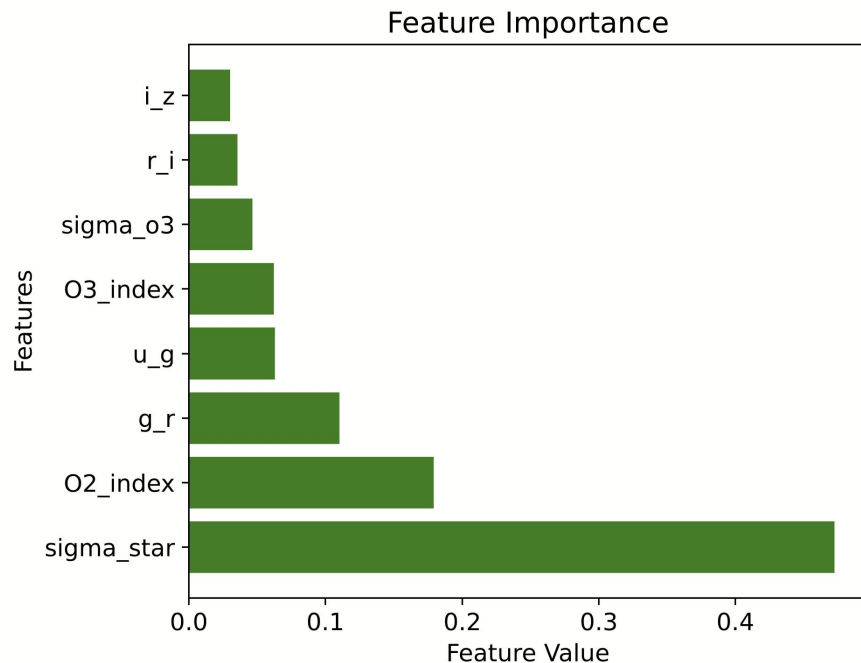


Finding the optimal values for four parameters of the XGBClassifier

Gradient-Boosted Model



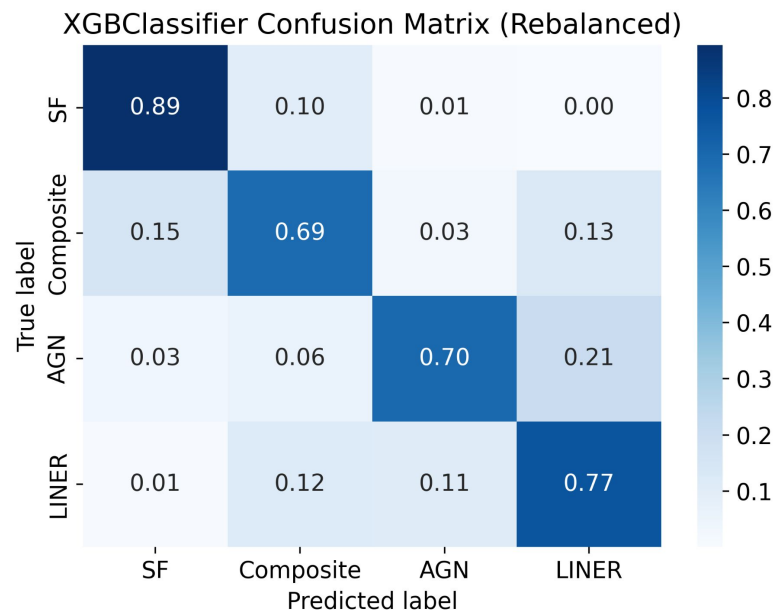
Gradient-Boosted Model



Gradient-Boosted Model

Adding manual thresholds to adjust decision-making:

SF	0.7
Composite	0.3
AGN	0.25
LINER	0.3

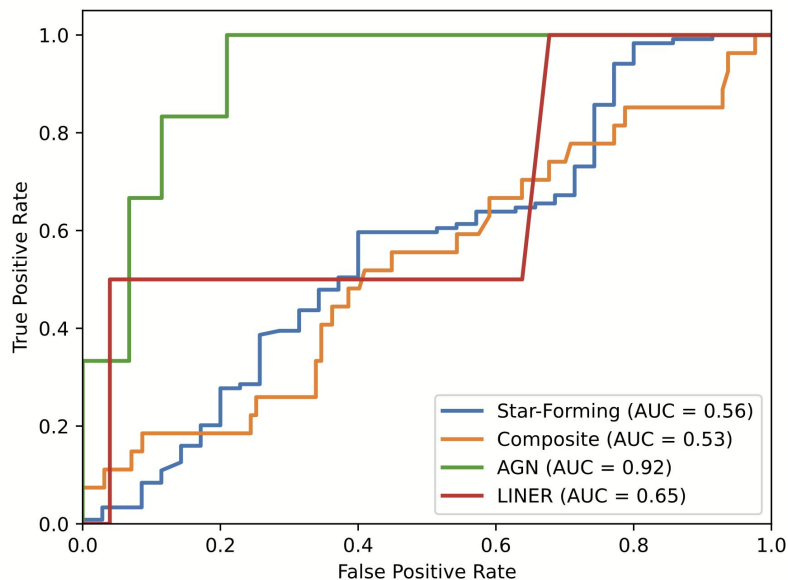


Intermediate Redshift Sources

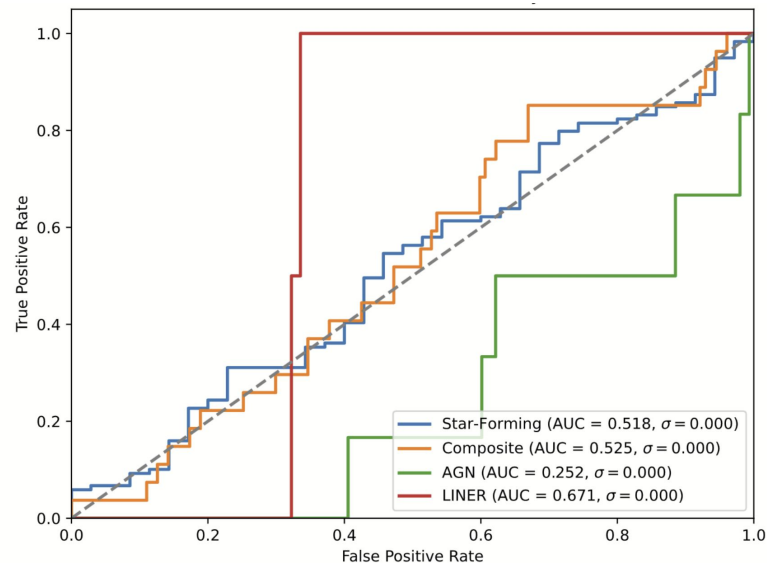
Acquired data from SDSS survey for $0.32 < z < 0.8$ that have been classified.

Our results are bad for **all** of our models

Intermediate Redshift Sources



XGBClassifier



Neural Network

Intermediate Redshift Sources

Acquired data from SDSS survey for $0.32 < z < 0.8$ that have been classified.

Our results are bad for **all** of our models

Possible explanations for this:

- Error in our data collection
- Features might not generalize well for higher redshifts
- Different data processing (but should not be a problem)

Conclusions

- Inconsistencies in the paper (Zhang et al., 2019) about data processing
- Improved the Balanced Accuracy & AUC scores (MLP & RF)
 - Different method to tackle class imbalance
- XGBClassifier also worked
- No good results for intermediate redshift sources
- Further work:
 - class balancing methods (e.g. **SMOTE** and/or **undersampling**)
 - obtain larger dataset



Extra slide: Optical diagnostic diagram

